Fiscal Year:	FY 2024	**Task Last Updated:**	FY 06/21/2024
PI Name:	Masson, Patrick H. Ph.D.		
Project Title:	Can Polyamines Mitigate Plant Stress Response under Microgravity Conditions?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	53706-1580	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	09/16/2019	**End Date:**	03/15/2024
No. of Post Docs:		**No. of PhD Degrees:**	0
No. of PhD Candidates:		**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/15/2024 per the PI and per NSSC information (Ed., 6/21/24). NOTE: End date changed to 09/15/2023 per NSSC information (Ed., 7/26/22).		
Key Personnel Changes/Previous PI:	July 2020 report: Dr. Shih-Heng Su is an Assistant Scientist in the Masson lab who has been contributing to this project since its inception.		
COI Name (Institution):	Su, Shih-Heng Ph.D. (University of Wisconsin-Madison)		
Grant/Contract No.:	80NSSC19K1483		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space exploration requires the design and implementation of bioregenerative life-support systems capable of maintaining breathable air, cleaning water, and producing food and fiber during space missions. Plants are expected to play key roles in these life-support systems. Therefore, it is important to understand the effects the microgravity environment has on plants to allow the design and development of better life-support systems for space exploration. Multiple studies have demonstrated that plants grown under microgravity display signs of stress, including altered morphology and expression profiles expected for stress response. Therefore, the development of mitigation strategies aimed at alleviating such stress will be important. Previous studies by multiple research teams including ours have demonstrated a role for polyamines in stress mitigation in plants. Polyamines are a group of compounds whose abundance increases in plants under stress. These signaling molecules have also been shown to contribute to stress mitigation in ground-based experiments. Therefore, we hypothesize that polyamines might also contribute to stress mitigation under microgravity, and propose to test this hypothesis by investigating the growth and expression profiles of Arabidopsis thaliana seedlings engineered to modify the pool of putrescine (one of these polyamines) accumulating in their tissues. Specific objectives will include: 1) Comparing the growth of Arabidopsis thaliana seedlings with genetic alterations affecting the pool of putrescine to that of wild type, within the VEGGIE hardware, both under microgravity on the International Space Station (ISS) and on the ground (1-g control), and 2) Use RNAseq approaches to investigate the effect of microgravity relative to 1-g ground control on the transcriptomes of wild type and mutant seedlings with alterations in their putrescine pool. These investigations should lead to a better understanding of the roles played by polyamines in stress mitigation under microgravity. It may identify novel genetic engineering methods that improve plant growth under microgravity, thereby allowing development of optimized bioregenerative life-support systems for long-term space exploration missions. This project may also have important applications in agriculture for crop production on marginal lands. Finally, our experiments will yield large sets of transcriptomic data from previously uncharacterized genotypes, which will be added to the GeneLab dataset.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project lead to a better understanding of the role of Putrescine and/or its derived signaling products in mitigating the stress associated with growth under microgravity environments such as those encountered during spaceflight and on the International Space Station. It also allowed the identification of novel genetic engineering strategies aimed at improving plant adaptation to the microgravity environment, with potential impact on our ability to improve the genome of other plant species / crops for utilization in bioregenerative life-support systems during spaceflight. A better understanding of plant responses to stress may also have important applications in agriculture for crop production on marginal lands. New large-scale RNAseq expression data associated with growth under microgravity for previously untested Arabidopsis genotypes were generated during this project and will be submitted to the GeneLab Data System. This project allowed us to introduce undergraduate students to the idiosyncrasies of scientific research in Space Biology, and also generated novel instructional materials allowing effective teaching of the difficult concepts of gravity perception by plants and their importance for agricultural production on Earth and during spaceflight. Such materials should be usable in K-12, undergraduate, and graduate education, as well as general-public outreach.
Task Progress:	The main objective of Advanced Plant Experiment (APEX-08) was to evaluate the impact of genetic alterations of the putrescine pool in Arabidopsis thaliana on its seedlings' ability to grow, develop and respond to the microgravity conditions of the International Space Station (ISS) relative to ground control. To this end, we developed and characterized Arabidopsis thaliana mutant and transgenic lines with alterations in putrescine metabolism including: (1) wild type Col-0, (2) adc1-1 knockout mutant affected in its ability to synthesize putrescine; (3) Cuao3-1 and cuao3-100cdr7 allelic mutants affected in their ability to degrade putrescine; (4) transgenic plants over-expressing the Poncirus trifoliata ADC (PtADC) gene involved in putrescine synthesis (hereafter named OXPtADC), and (5) transgenic cuao3 plants over-expressing the AtCuAO3 transgene (hereafter named OXCuAO3). The PtADC over-expressing plants and Cuao3 knockout mutants were shown to accumulate more putrescine in shoot and root tissues than the other lines, allowing us to test the potential effect of increased putrescine content, or related metabolic changes, on plant responses to the stress associated with spaceflight conditions. To evaluate the effect of alterations in the pool of putrescine and/or its catabolites on Arabidopsis thaliana seedling responses to the stress encountered during spaceflight, we germinated Arabidopsis seedlings of all six genotypes in petri dishes with the APEX growth chamber on ISS, and ran a similar control experiment on the ground, at NASA Kennedy Space Center, mimicking the growth conditions measured on ISS. At the end of this growth period, the plates were photographed for subsequent analysis of seedling growth. The seedlings were harvested, fixed in RNAlater® and returned to the laboratory for RNA extraction and expression analysis. Results indicate that seedlings of all genotypes grown under microgravity conditions on ISS carry longer leaf petioles than those grown in a ground control at NASA Kennedy Space Center (KSC). This difference was less pronounced in putrescine over-expressing lines (OxPtADC and cuao3 mutant) than wild type. Additionally, the roots of ISS-grown seedlings were oriented more randomly than those of ground-control seedlings. To characterize the effects of spaceflight on the patterns of gene expression, we dissected root and shoot tissues from RNAlater-fixed seedlings of all genotypes, extracted RNA from these samples, and sequenced them. The root and shoot tissues of all genotypes displayed strong gene-expression responses to the spaceflight conditions, with expression responses preferentially associated with hypoxia, reactive oxygen species (ROS) signaling, cell wall metabolism, and photosynthesis, amongst other things. Importantly, plants over-expressing the PtADC gene and cuao3 knockout plants displayed dramatically simplified transcriptional response profiles to spaceflight relative to ground control, suggesting a role for putrescine accumulation in plant tissues on their ability to tolerate the stress associated with spaceflight conditions. Considering this important observation, we also set up to investigate the ability of plants with the same genetic alterations in the polyamine metabolic pathway to grow and develop on lunar regolith simulant (LHS-1), a substrate previously suggested to be stressful for plants. The results of preliminary seedling-viability experiments suggest plants over-expressing the PtADC transgene also display increased tolerance to lunar regolith simulants. Taken together, our results suggest engineering plants for increased accumulation of putrescine within their tissues upon exposure to microgravity and/or lunar regoliths may confer improved tolerance to these novel environments. This observation may have important implications for the design of plant-based bioregenerative systems aimed at supporting

	life in long-term space exploration missions and/or exoplanet colonization programs.
Bibliography Type:	Description: (Last Updated: 07/03/2024)
Abstracts for Journals and Proceedings	Masson PH, Su S-H, Keith M, Kinney F. "Using genome-wide association studies and RNA-seq to investigate the molecular mechanisms underlying root circumnutation in Brachypodium distachyon." PAG 31, International Plant and Animal Genome Conference, San Diego, CA, January 16, 2024. Abstracts. PAG 31, International Plant and Animal Genome Conference, San Diego, CA, January 16, 2024. Abstract: 52206. https://pag.confex.com/pag/31/meetingapp.cgi/Paper/52206 , Jan-2024
Abstracts for Journals and Proceedings	Su S-H and Masson PH "Alterations in the Polyamine Metabolic Pathway Help Alleviate the Stress Associated with Plant Exposure to Spaceflight Conditions. " Seminar Seminar, Department of Biology, Texas State University, San Marcos, TX , Apr-2024
Awards	Su S-H. "University of Wisconsin-Madison Award for Mentoring Undergraduates in Research, Scholarly and Creative Activities." May-2023